

Introduction

In the realm of auditory neuroscience, traditional fMRI studies rely on controlled, artificial stimuli such as isolated tones and snippets of speech to map the brain's auditory network. In natural contexts however, the human ear and auditory cortex has to continuously parse complex and overlapping signals. With Nastase et. al. (2021)'s release of the Narratives dataset, we have access to a rich corpus of listener's neural activity in response to naturalistic stimuli. From this set, I chose "The Tunnel Under the World," a short story by Frederick Pohl that was aired on the NBC radio with accompanying music and sound effects.

My primary research question is: do time-varying acoustic features of narrative audio, including sound energy, spectral structure, and pitch, predict patterns of brain activity during naturalistic story listening? Furthermore, are these neural responses to acoustic features different during segments containing just speech, just music, or both speech and music? By extracting these features using analysis libraries such as librosa that I can align to the audio and neural timeline, I can run various classification and ISC analyses to explore how the brain encodes these physical characteristics of sound, beyond the actual content of the speech that is being heard. I also plan on manually tagging the audio into three categories: speech-only, music-only, and speech + music, which can be used as grouping variables. This will allow the differential analysis of how listeners' brains encode information under these conditions.

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2. Setup

Below, we import all the necessary libraries for the notebook and set up paths to directories that we will pull data from.

```
Dir /tmp/final_project_results/ created.
```

2A. Load Audio

We use the `librosa` library to load the Tunnels Under the World audio. It decodes the file and returns a time-series of floats (using the default sampling rate of 22.05 Hz). This usually does not take too long.

```
Loaded /home/NEU480/datasets/narratives/stimuli/tunnel_audio.wav
Duration: 1533.77 seconds
Native SR: 22050
Time to load: 2.54 seconds
```

2B. Extract Sonic Features

We begin by extracting **sound energy root mean squared** and **spectral flux**. The former measures the effective amplitude of the sound signal, i.e. its power or loudness, by calculating the root of the mean of adjacent instantaneous values, squared. The latter detects how quickly a signal's power spectrum changes over time, thereby capturing when sounds are onset and with what strength.

	time	rms_energy	spectral_flux
0	0.00000	0.001695	0.000000
1	0.02322	0.001705	0.000000
2	0.04644	0.001722	0.000000
3	0.06966	0.001150	0.121475
4	0.09288	0.021806	0.017866
5	0.11610	0.054559	0.130707
6	0.13932	0.067246	4.359818
7	0.16254	0.087558	9.529262
8	0.18576	0.114422	4.745287
9	0.20898	0.137655	1.467012

Some other features we are looking at include **spectral centroids** and **Mel-Frequency Cepstral Coefficients (MFCCs)**, which inform our understanding of the audio's spectral structure. The centroids, per their name, indicate where most of the sound's energy is concentrated within its spectrum and measures the perceived "brightness" or "timbre" of the sound. MFCCs attempt to mimic how ears perceive sound (and therefore how the cortical region integrates diverse sonic structures) across 13 variables, where the first captures sonic energy, 2-4 the broad shape of the spectrum, and 5-13 various finer details from harmonies to consonants.

Lastly, we also extract **pitch**, important for both speech and music. We take the fundamental frequency from the audio, which will be used later for aligning with neural data from regions that may selectively respond to pitch.

Note: This steps tends to take ~10 minutes to complete.

Time to load: 887.50 seconds

Now, we combine and time-align everything by converting from real time to TRs, using the dataset-proposed TR of 1.5. For the potentially multiple frames that fall within a single TR, we take their mean.

Before we begin neural alignment, we want to redefine the sound energy column. The human ear perceives loudness logarithmically, so we will convert the linear RMS energy into a decibel (dB) scale to better represent auditory perception. To avoid $\log(0)$ issues when there's pure silence, we add a miniscule constant.

2C. fMRI Preprocessing

While brain data preprocessing steps have already been taken, such as motion correction, scanner noise, etc., we still have to convolve our audio data to the brain. Specifically, whereas `librosa` notes a potential peak in sound energy instantaneously, the BOLD fMRI signal peaks ~5-6 seconds after that same peak. Therefore, we need to convolve the sonic features with a hemodynamic response function that simulates the brain's delay, which allows us to properly compare the two.

We standardize our values using Z-scoring. Note that features like F0 (Hz) and Sound Energy (dB) have different scales, and later regression may unfairly weight features simply because of larger raw numerical values.

The last thing we want to do for alignment is to note the 23 seconds of silence after the stimulus where the scanner kept running. Since that is present in the brain data, making up

a total of 1040 TRs, we need to pad this dataframe with 17 TRs, since the audio was 1534 seconds long in comparison to the 1560 seconds of scanner runtime.

2D. Tagging

As our last step of preprocessing, we want to enable experimental analysis on our "speech" vs. "music" hypothesis. This means we have to tag the audio with TR-aligned labels that correctly represent speech, music, or both.

A hybrid approach will be used:

1. `inaSpeechSegmenter` - a convolutional neural network-based library that splits audio into speech, music, and noise. It outputs a table in the format (label, start, end), where the latter two fields are in seconds. However, this does not natively output a 'both speech & music' label.
2. `Gentle` - a time-aligned transcript with word-level start and end labels, also in seconds. This can capture speech labeling.

So, a 'both speech & music' label can be created by overlapping the two outputs.

Note: `Segmenter()` fails to run on Adroit because it does not come installed with its neural network weights to save space and attempts to download it from a URL. Since Adroit is offline, open a login node that has internet access and paste the following line to cache the weights in your home directory:

```
python -c "from inaSpeechSegmenter import Segmenter; Segmenter()"
```

This also takes quite a while (~20 min) to run.

```
E0000 00:00:1778692617.795387 36040 cuda_platform.cc:52] failed call to cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)
```

```
1997/1997 - 42s - 21ms/step
```

```
1596/1596 - 168s - 105ms/step
```

```
Time to load: 2401.99 seconds
```

	label	start	end
0	male	0.00	2.42
1	noEnergy	2.42	3.20
2	male	3.20	4.80
3	noEnergy	4.80	5.38
4	music	5.38	13.02
	label	start	end
4	music	5.38	13.02
29	music	66.82	78.54
43	music	121.92	124.94
45	music	134.10	135.58
86	music	209.00	215.34

Unfortunately, though `inaSpeechSegmenter` did a great job at detecting speech and music segments separately, it performed much worse at detecting when they occurred

simultaneously. This is partly due to the Tunnel audio's format: long dialogue sections are punctured by short bursts of orchestral incidental music, which usually feature for 3-4 seconds before fading rapidly in volume and playing at a very low decibel as the narrator or a character starts talking. They usually only co-occur for a few seconds before the music fades out completely, so due to this constraint we chose to forgo analysis of the 'both speech and music' label. However, we still overlap the outputs of the two engines to create a comprehensive TR-by-TR capture of the speech and music of the audio. Combining the inaSpeechSegmenter with Gentle also helps to smear any noise either may have and validate the speech-music boundaries.

Gentle transcripts have a specific JSON format that we will parse now. Additionally, as the dataset metadata mentioned 126 unsuccessfully localized words, this will also show in the transcript, specifically in the `case` field as `not-found-in-audio`. Looping through every word looking for the `start` and `end` fields will throw a key error for those words, so we begin by filtering them out.

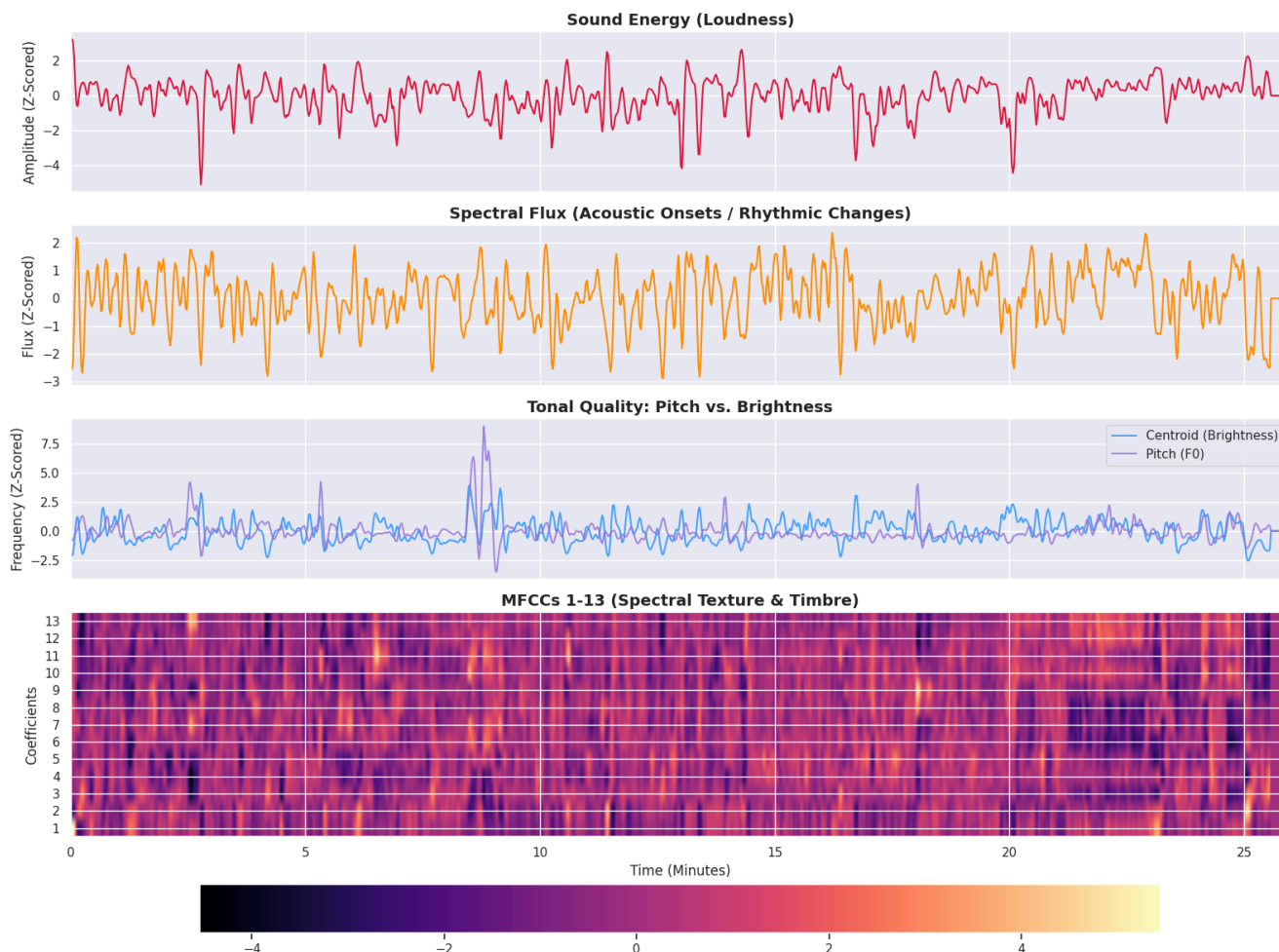
```
Total words: 3435
Successfully aligned speech blocks: 3309
```

Now we can align the inaSpeechSegmenter data with the Gentle transcript.

```
Tagging complete. Final matrix shape: (1040, 24)
  tr_index category  is_speech  is_music  is_speech_music
0         0  speech      True    False             False
1         1  speech      True    False             False
2         2  speech      True    False             False
3         3   music     False     True             False
4         4   music     False     True             False
```

3. Initial Testing

We can finally begin our running analyses. We start with some basic visualizations to see the sonic structure of the Tunnel audio. Note that MFCCs are dimensionless numerical values.



3A. Encoding

Let's try aligning these features to the brain. We will focus primarily on the auditory cortex in the superior temporal lobe (Heschl's Gyrus) and the superior temporal gyrus. To get the BOLD signals for these specific ROIs, we align the brain to the Harvard-Oxford cortical atlas and extract those patches. For now, let's work with just one subject to confirm some preliminary results to assure us of how we parsed the audio data.

`[fetch_atlas_harvard_oxford]` Dataset found in `/home/NEU480/datasets/nilearn_c`
Heschl's Gyrus index: 45
STG Anterior index: 9
STG Posterior index: 10

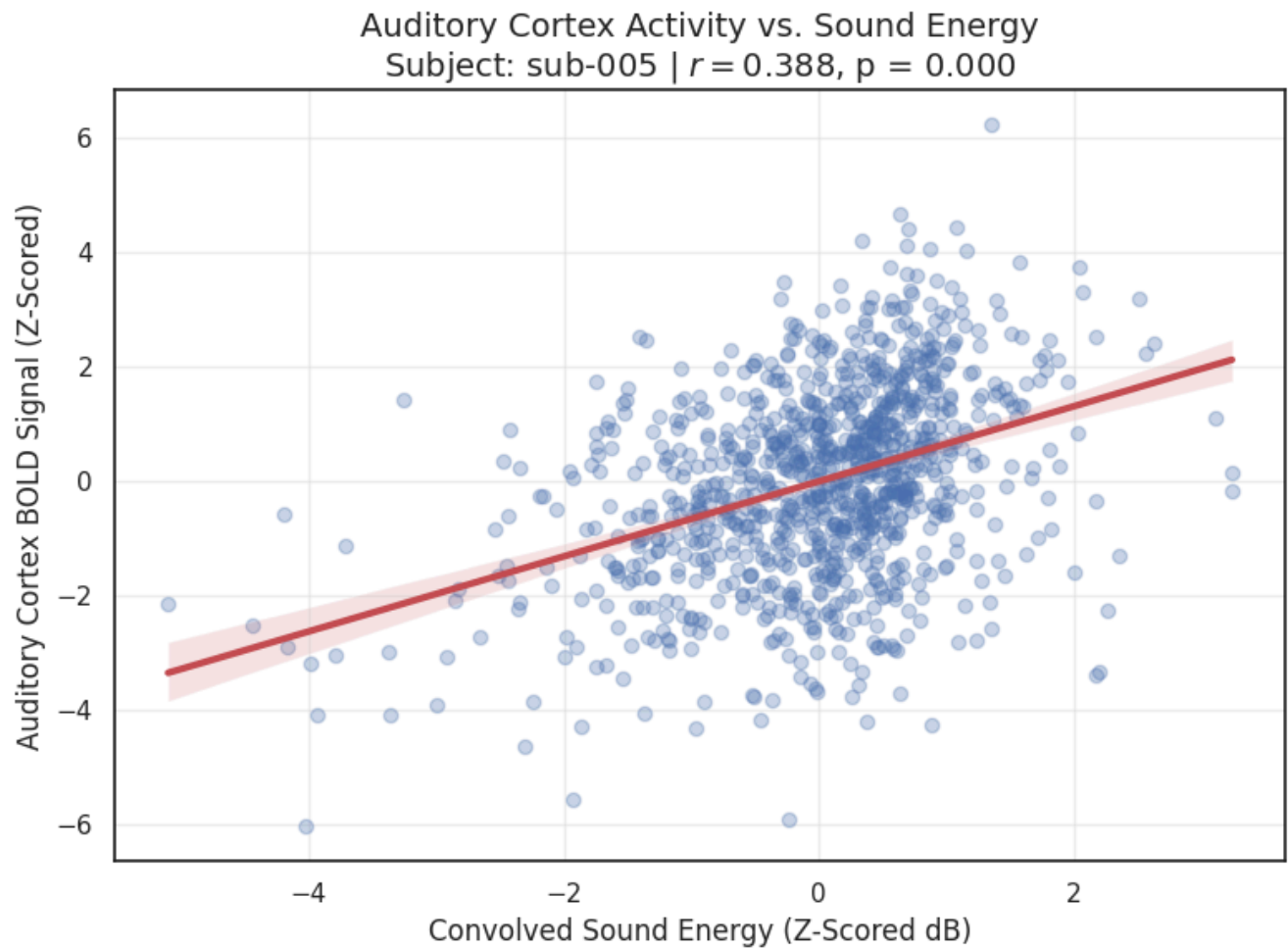
Processing subject: `/home/NEU480/datasets/narratives/derivatives/afni-nosmooth/sub-005/func/sub-005_task-tunnel_space-MNI152NLin2009cAsym_res-native_desc-clean_bold.nii.gz`

Extracted time series shape: (1040, 48) (Expected: 1040 TRs, 48 Regions)

Our first experiment is to correlate sound energy with BOLD activity. Schmalzle et. al. (2022) found that in naturalistic audio listening, temporal changes in sound energy predicted moments of heightened attention and synchronicity. While we explore

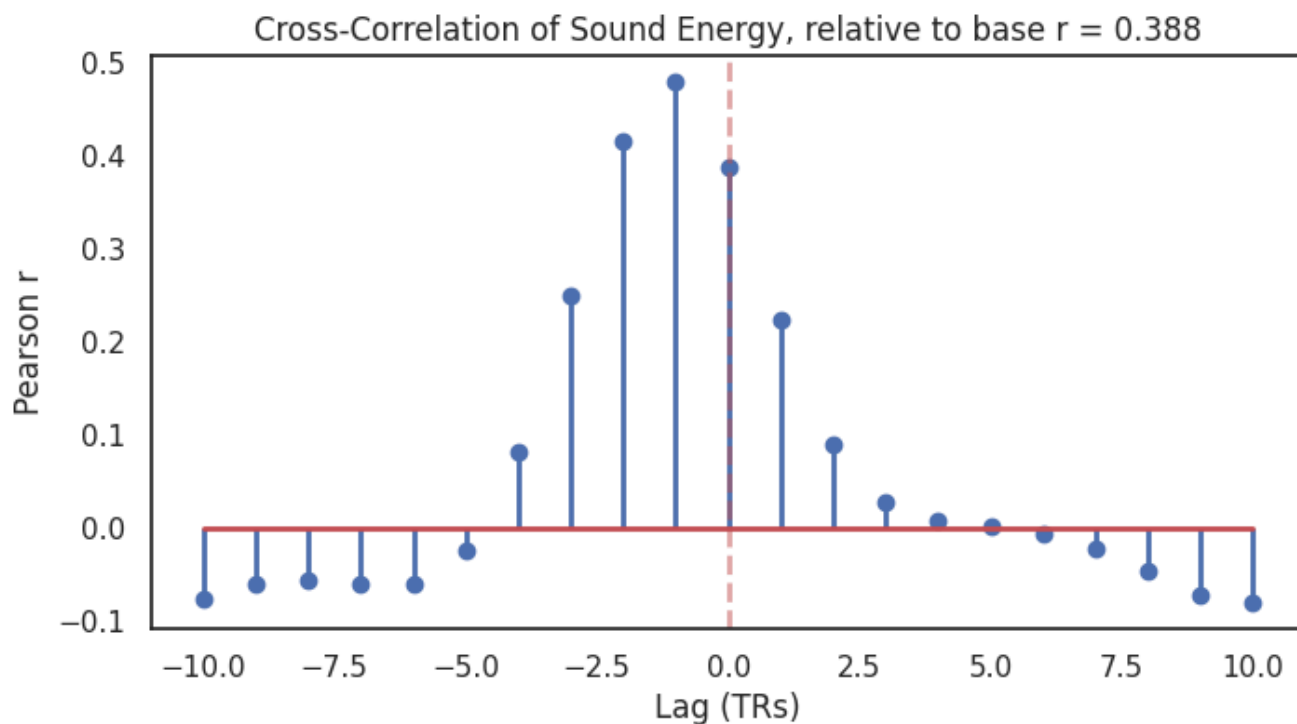
synchronicity later using ISC, we test here if there is indeed a positive correlation between energy and brain activity.

Correlation (r): 0.3881
P-value (p): 9.9751e-39



This is a good result! With a r -value of 0.3881 and a significant p -value, we have shown that there is a strong relationship between sound energy and a participant's brain activity in the auditory cortex. However, in another preliminary test I ran (not included in this notebook), I had to measure cross-correlation, i.e. the correlation between two signals if one was displaced relative to the other, and found the following results:

Base Correlation: $r=0.3881$, $p=9.9751e-39$



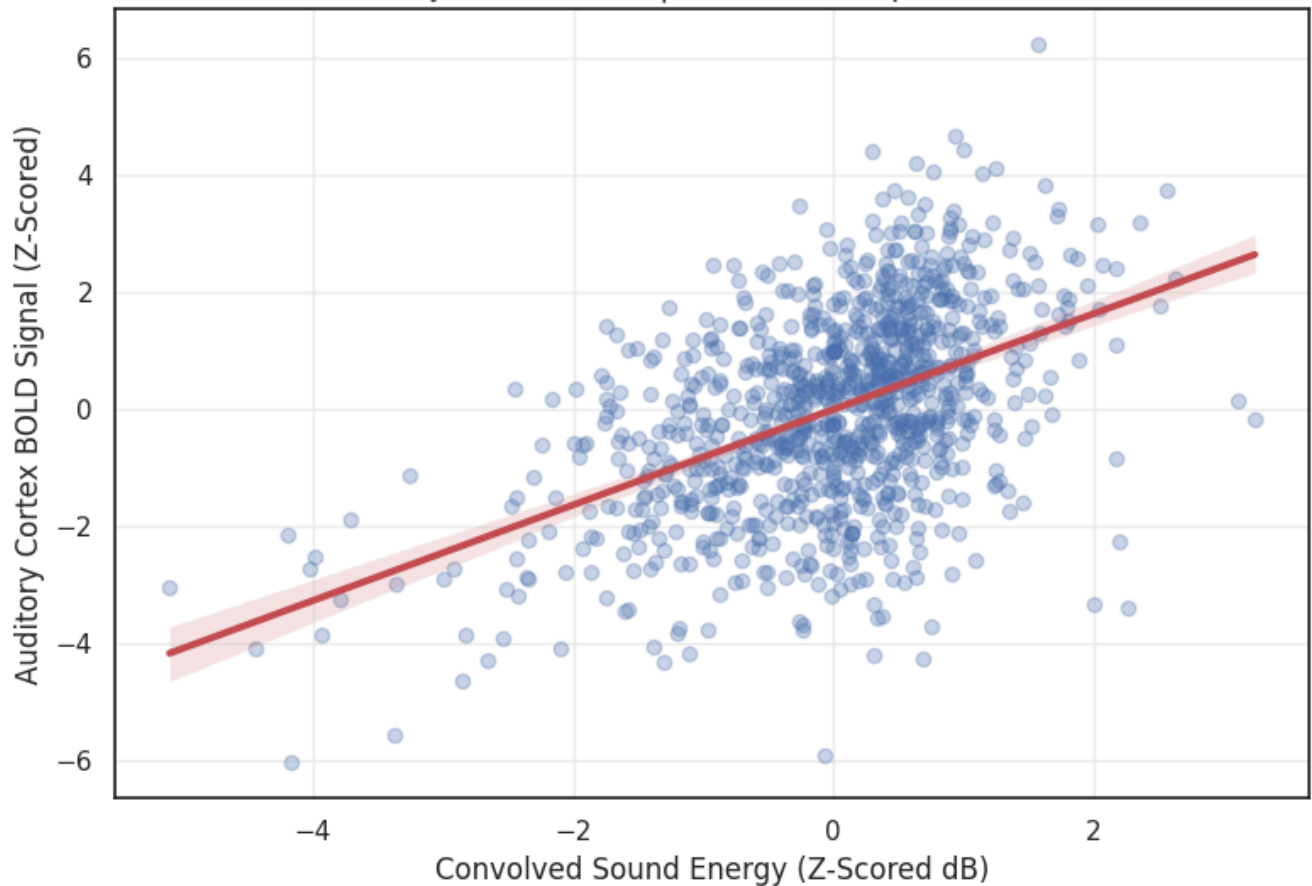
While this does follow the general HRF basis function that we might expect to see, with decreasing correlation in either direction of the peak, it seems that the time alignment preset by the dataset may have been too much. In investigating this phenomenon, I came across Handwerker et. al. (2004), which found that hemodynamic responses varied across subjects and crucially, brain ROIs. Time-to-peak or time-to-onset had an observed 2.5 seconds window of misestimates. The general HRF convolution assumes an averaged response time for the entire brain, but with this differential HRF, it could be the reason why the auditory cortex's correlation seems to peak an entire TR earlier. To capture this important characteristic, we need to temporally shift everything 1 TR back.

Applied -1 TR physiological shift. Final shape: (1040, 24)

With the correct temporal padding, let's try the previous sound energy vs. activity in Heschl's Gyrus again.

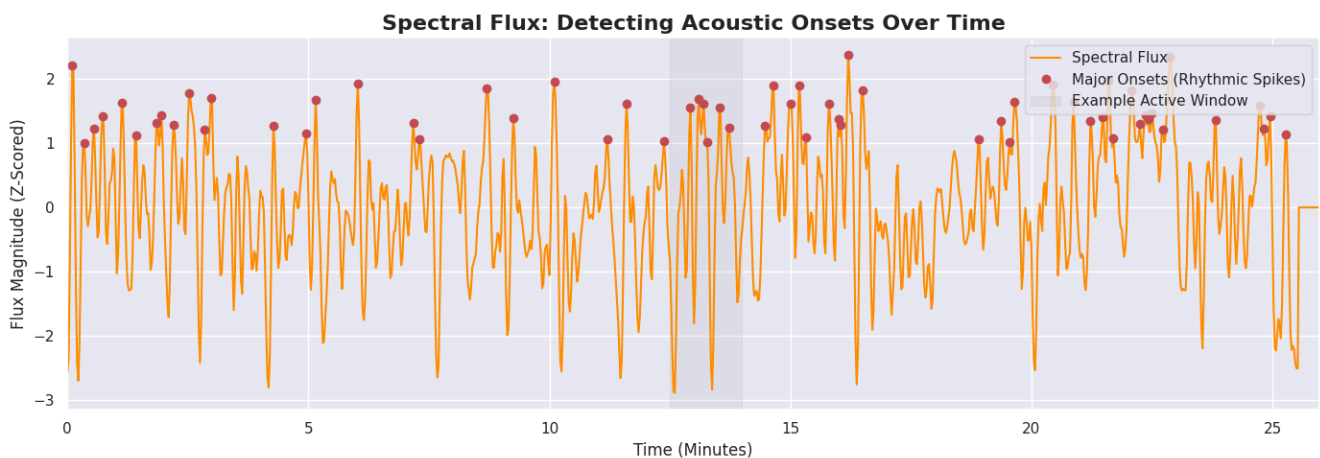
Corrected Correlation: $r=0.4807$, $p=3.0273e-61$

Heschl's Gyrus Activity vs. Sound Energy
Subject: sub-005 | $r = 0.48071$, $p = 0.000$



Much better! Our correlation went from 0.3188 to 0.4807, a huge increase. We will use this correctly aligned data for the rest of the analyses, but this also shows why it is important to verify previous results. Though we did have a positive correlation, we expected a slightly stronger relationship, which we now have.

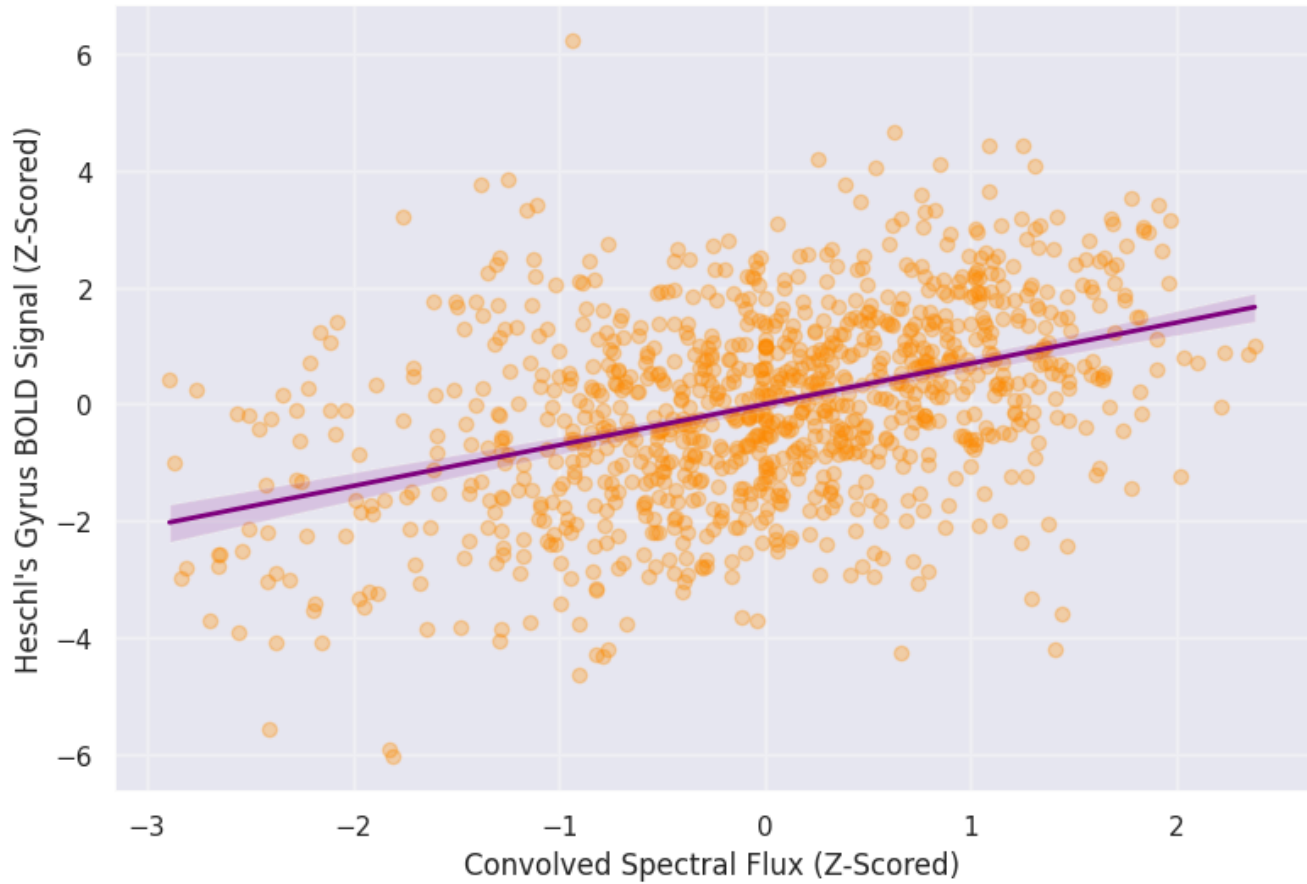
Now, let's do a similar analysis for spectral flux.



Overall A1 Correlation with Spectral Flux: $r=0.4125$, $p=5.5105e-44$

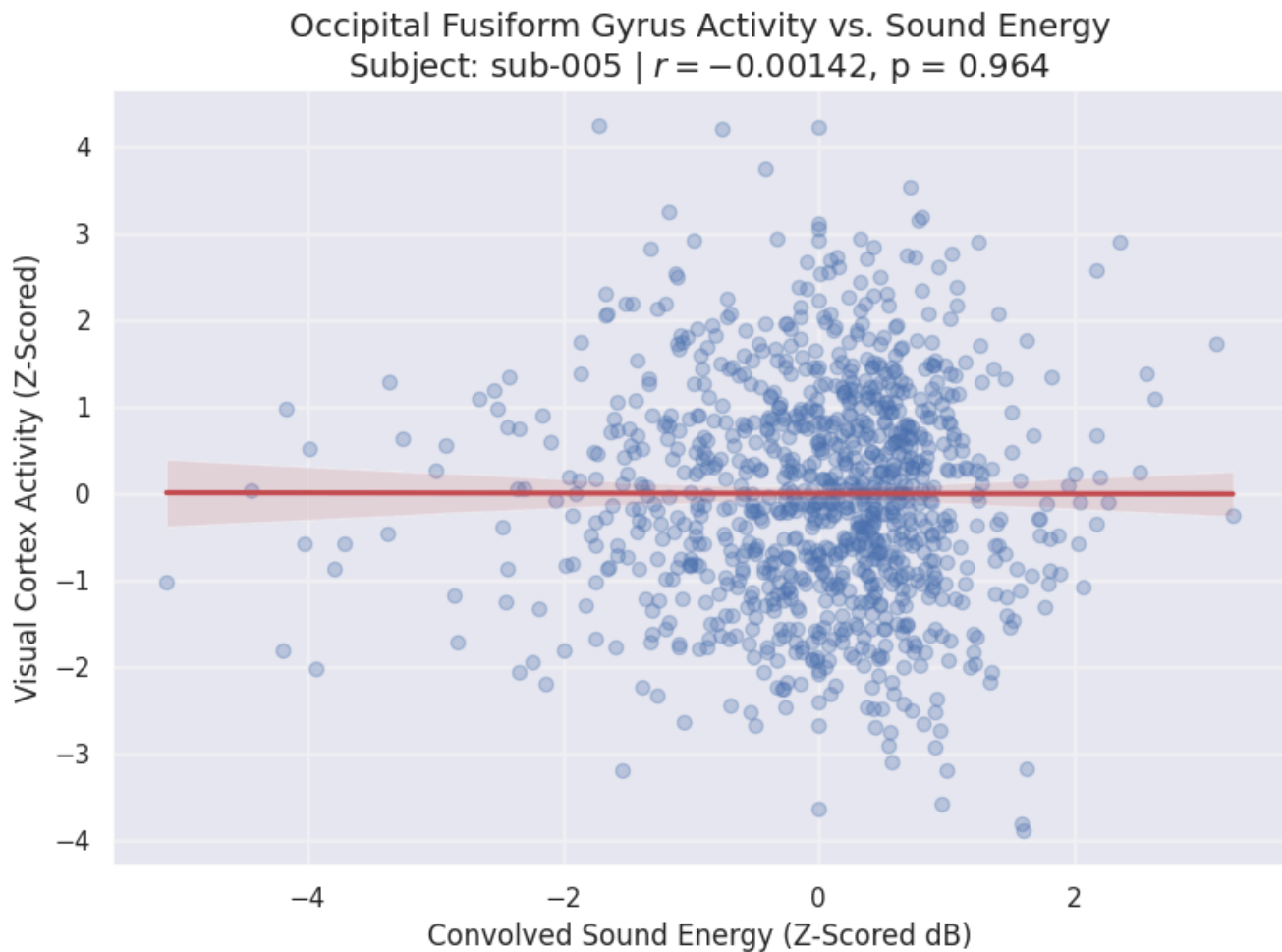
Heschl's Gyrus Activity vs. Spectral Flux

Subject: sub-005 | $r = 0.412$



One final check we'll do is to run the sound energy analysis on a control ROI. Here, we pick the Occipital Fusiform Gyrus, a key area of the visual cortex but not of the auditory system, in theory. We expect that it should have very little correlation with sound energy.

Corrected A1 Correlation: $r = -0.0014$, $p = 9.6361e-01$



Indeed, that's what we find! An almost horizontal line with r-value -0.00142 and p-value 0.964. With this initial confirmation of the validity of our audio parsing, we move on to novel analyses.

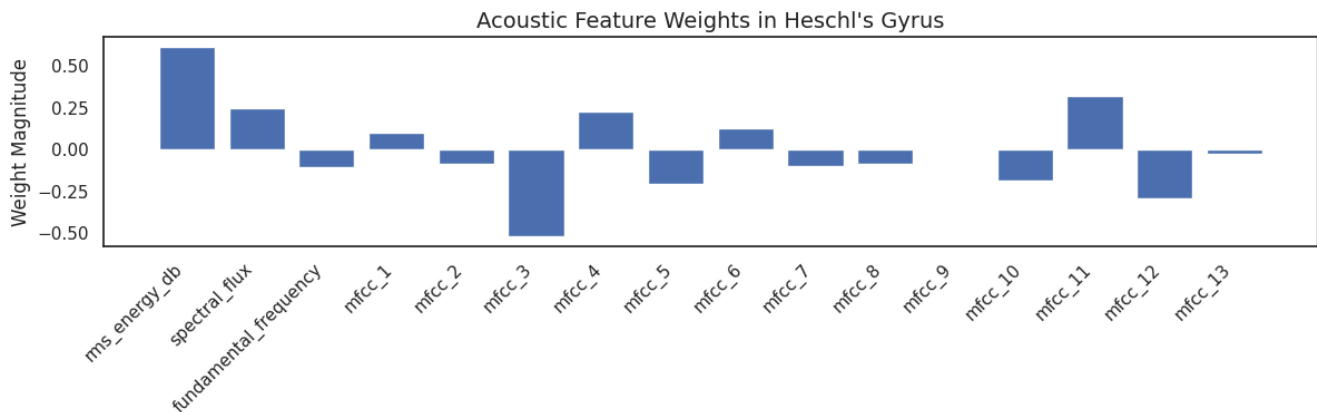
4. Classification

4A. Encoding Part 2

We want to see how well we can predict the brain's activity based on the continuous physical properties of the sound. Since we expect quantities such as sound energy and spectral flux to rise and fall together, we use L2 regularization as they are highly correlated. We first split into training and testing splits with a 80/20 split, standardize the training features, then cross-validate to find the best performing penalty parameter before testing it on the held-out data.

Encoding Model R^2 Score: 0.4365
Best Alpha: 5.99

The model score is not bad, but to see which features are driving the primary auditory cortex activity the most, we plot them below and see which weights (betas) are largest.



Unsurprisingly, sound energy and spectral flux seem to be the largest impact features, though it seems mfcc_3 also has a strong negative weight. MFCC3 generally captures mid-frequency peaks and valleys, typically associated with formant structures of human speech such as differentiating vowels. Since Heschl's gyrus does capture a lot of the finer details of human speech, this relationship makes sense.

In thinking about components such as fundamental frequency, we probably would not expect there to be a strong linear relationship between it and BOLD activity. That would imply that higher notes inspire higher activity overall, which intuitively does not make sense even as there may be brain regions that selectively respond to certain pitches.

4B. SVM Classification and Decoding

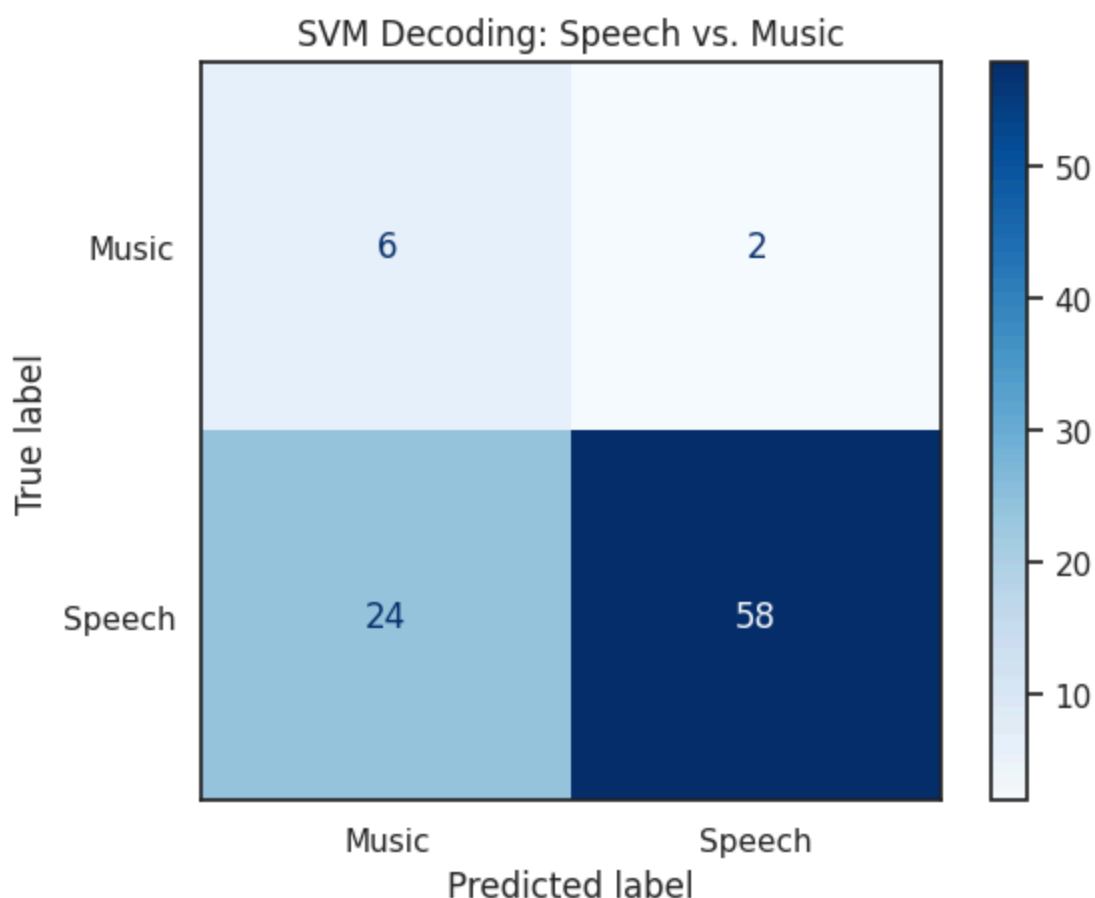
For this analysis, we want to go the other direction: given brain activity, can we decode the sound that the participant is listening to? To focus this question, we look at the category label: speech or music (or silence). Earlier, we added binary indicator columns to each TR that represented whether it contained speech, music, neither, or both (which we are ignoring). So, we look at the TRs that are associated with music or speech only, retrieve the data and labels associated with such, and run our classification model.

We use support vector machines (SVM) with a linear kernel since this is a high dimensional space both for brain data and audio data, and this will hopefully avoid overfitting. We also use grid search to cross validate for a cost parameter, then fit it to the testing set to test performance.

Total TRs for decoding: 900
 Class balance (Speech=1, Music=0):
 1 815
 0 85
 Name: count, dtype: int64

Best C: {'C': 0.01}

	precision	recall	f1-score	support
Music	0.20	0.75	0.32	8
Speech	0.97	0.71	0.82	82
accuracy			0.71	90
macro avg	0.58	0.73	0.57	90
weighted avg	0.90	0.71	0.77	90



Unfortunately, the results above say more about machine learning models than genuine brain-audio decoding. With a large imbalance between the two labels, with 85 for Music and 815 for Speech (roughly 10:90 split), the model could guess Speech every time and have 90% accuracy. Even with `classweight='balanced'` in the SVM, which adjusts weights inversely proportional to sample sizes, the performance is still not great. We can see how though precision for Speech is 0.97, the recall is 0.71, meaning it only found 71% of the actual speech, with the other 29% was classified as music. On the other hand for Music, the recall score is great since capturing 75% of only 8 music TRs amongst the high dimensional data demonstrates success, but the low precision rate of 20% shows that the

model guessed Music frequently, with a lower threshold for deciding what was truly Music, and was incorrect most of the time. While the brain's activity pattern for music may be distinct enough to be picked up on, this is not a significant result.

Before moving on, we do want to try downsampling the number of Speech TRs to 85 as well so there would be an even split between either label. Chance accuracy would then sit at 50%, and any score higher than that may be significant to our findings. We do this by extracting the speech and music TRs into separate dataframes, then using `sklearn.utils.resample` to scale down the speech dataframe. The rest of the SVM training and testing procedure remains the same.

```
New Class Balance (Speech=1, Music=0):
```

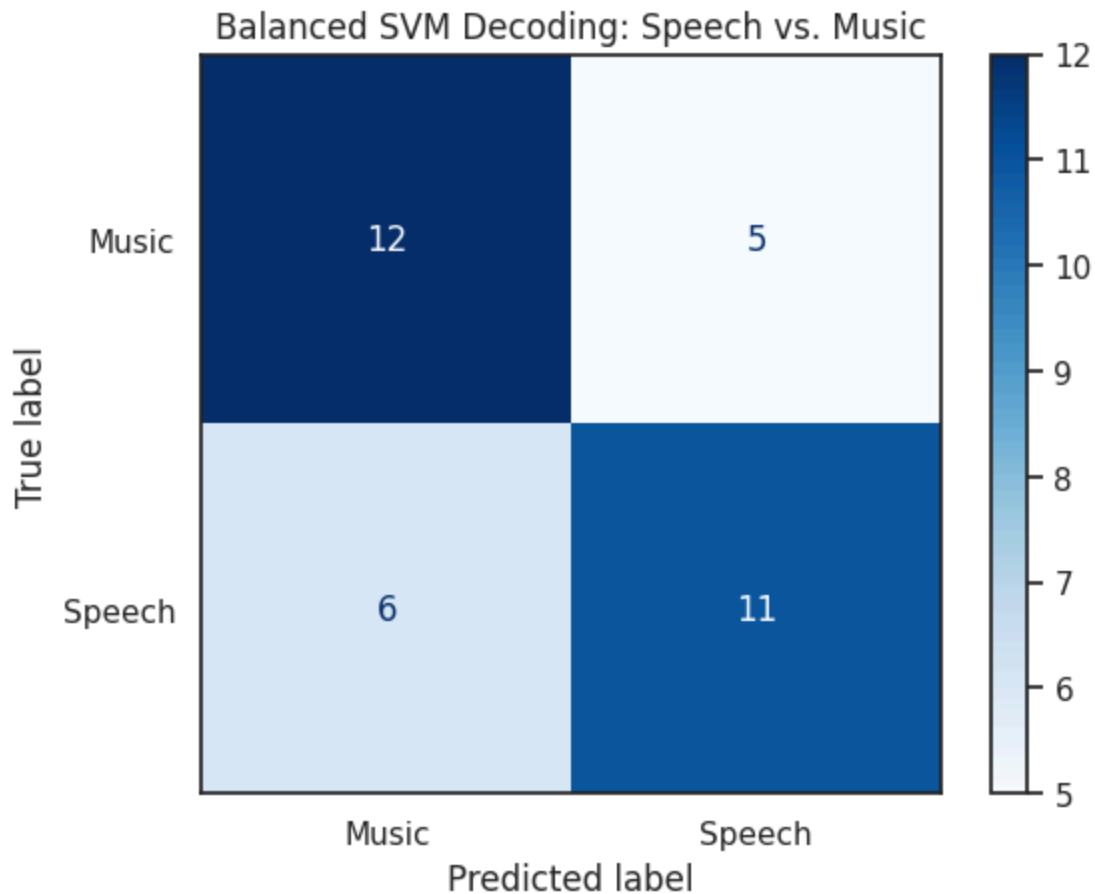
```
0      85
```

```
1      85
```

```
Name: count, dtype: int64
```

```
Best C: {'C': 0.01}
```

	precision	recall	f1-score	support
Music	0.67	0.71	0.69	17
Speech	0.69	0.65	0.67	17
accuracy			0.68	34
macro avg	0.68	0.68	0.68	34
weighted avg	0.68	0.68	0.68	34



Downsampling helped a lot, with better precision and recall scores across both Music and Speech that are comparatively similar. We were able to secure 68% accuracy, which is much higher than chance. While this can be extended to cover other parts of the auditory cortex and across multiple subjects, this initial result suggests that the brain does seem to differentially respond to music and speech.

5. ISC

We move from one subject analysis to multi. This will allow us to answer questions regarding synchronicity. Schmalzle et. al. (2022) previously found that across participants there was heightened attention during loud (high sound energy) moments. We want to see if that is true in this dataset, and if there are any other features that have high ISC.

The eventual format we want the data to be in for BrainIAK is (TRs, Voxels, Subjects). So we load in each subject at a time and fit our Harvard-Oxford atlas masker to their data so we can start calculating ISC. We first look at `participants.tsv` to select the participants that actually listened to Tunnel, then excluded the ones that the dataset mentioned have an incorrect number of TRs.

Found 23 subjects assigned to the 'tunnel' task in metadata.
Proceeding with 21 valid subjects after exclusions.

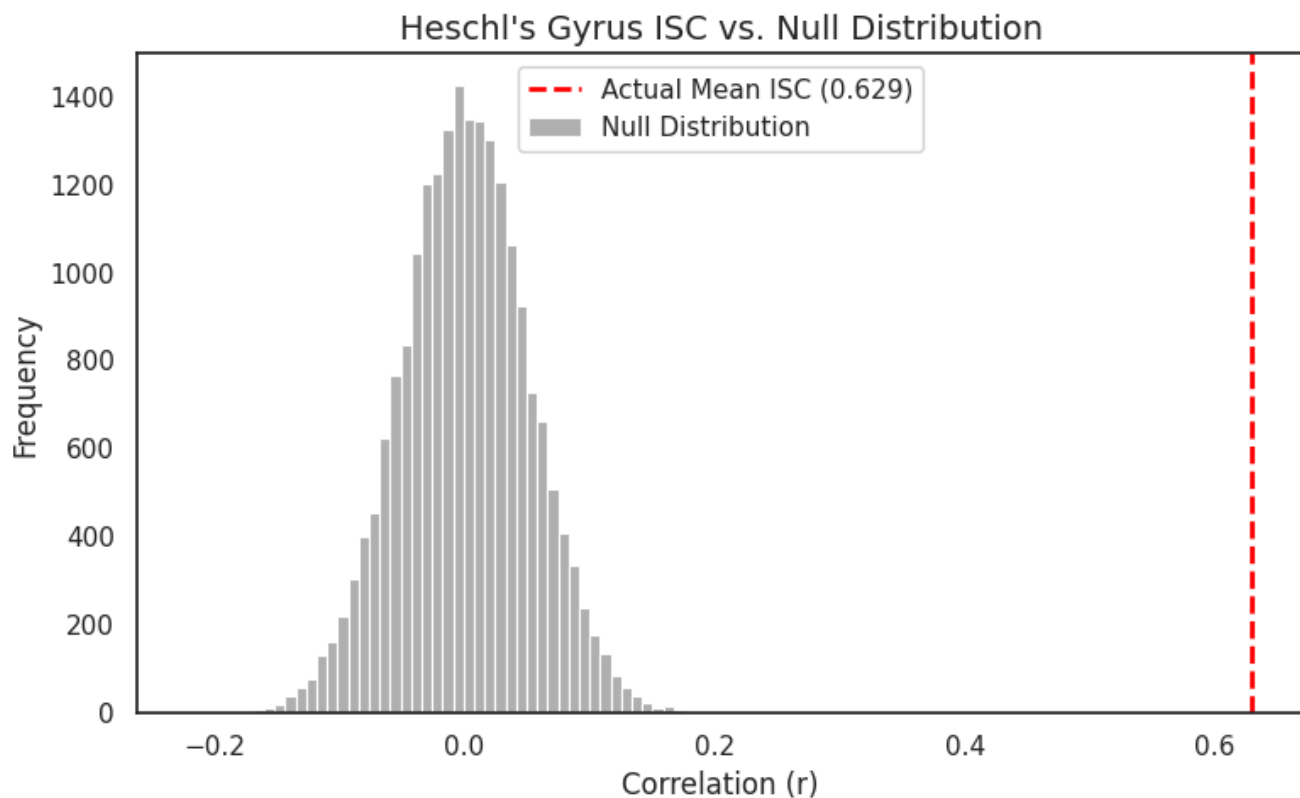
```
Loading data for sub-001.
Loading data for sub-005.
Loading data for sub-022.
Loading data for sub-025.
Loading data for sub-026.
Loading data for sub-041.
Loading data for sub-052.
Loading data for sub-053.
Loading data for sub-055.
Loading data for sub-056.
Loading data for sub-057.
Loading data for sub-058.
Loading data for sub-059.
Loading data for sub-060.
Loading data for sub-061.
Loading data for sub-062.
Loading data for sub-063.
Loading data for sub-064.
Loading data for sub-065.
Loading data for sub-066.
Loading data for sub-069.
Data loaded for all participants.
Final ISC array shape: (1040, 48, 21)
```

With the data all loaded, let's start by calculating overall ISC in the auditory cortex across participants using LORO.

```
Mean ISC in Heschl's Gyrus: 0.6292
```

We see a high ISC value, but we need to test its statistical significance. We can do so by creating a null distribution by randomizing the order of the data and comparing it to the actual data.

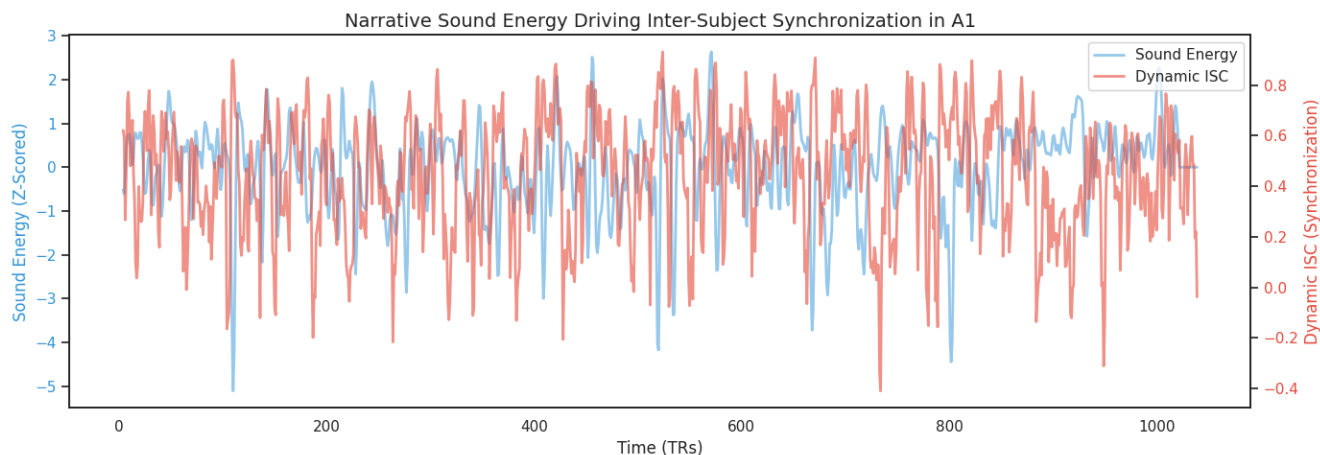
```
p-value relative to null distribution: [0.61938062 0.11388611 0.77222777 0.12987013
0.38961039 0.20679321
0.8021978 0.04995005 0.59140859 0.54845155 0.18781219 0.73026973
0.68231768 0.00699301 0.03596404 0.04295704 0.36663337 0.78621379
0.08591409 0.01398601 0.78821179]
```

Honestly--I am not sure why the p-values are all over the place, with some significant and others not, when the average ISC mean is very high and has no plotted overlap with the null distribution. Further analysis would have to be done into individual participants to see if there is noise in those participants, or if something is wrong with my code.

We move on to checking how ISC modulates in relation to the trajectories of our sonic features. Our approach uses sliding windows of 7.5 seconds (5 TRs) to capture the mean ISC per window and align it to individual features.

Calculating ISC across 1036 windows.



To support this plot, we also did peak analysis to see if there was a verifiable difference in ISC between moments of low and high sound energy. Using `scipy.signal.find_peaks`, we

could locate the peaks along the sound energy curve and match them to the ISC values then. Conversely, we can also invert this and find the nonpeak moments to get a baseline for comparison. We then run a t-test to see if the difference is significant.

Number of Sound Energy Peaks identified: 57
Mean ISC during Audio Peaks: 0.5150
Mean ISC during Baseline Audio: 0.4421
T-Test Results: $t = 2.42$, $p = 0.0183$

With a mean ISC different of ~ 0.07 between moments of audio peaks and baseline non-peak sound energy, we found that there is indeed a significantly higher level of synchronicity amongst participants when there is high sonic energy.

Given that we also have the categorical labels set up, it is worth running an experiment to see if synchronicity is different when participants are listening to speech vs music. This can be done by simply taking the TRs associated with just music or just speech and running ISC on the activity associated with those TRs.

Group Synchronization during Music: 0.6421
Group Synchronization during Speech: 0.6236

It does not seem there is any significant difference in mean ISC between the two conditions except that Music is slightly higher. If the difference was significant, it might be because Music often occurred suddenly and loudly in the story and therefore may be a result of sonic energy and spectral flux.

5. DISCUSSION

Throughout this notebook, we have conducted a study of how the brain responds to naturalistic story listening with a specific lens of auditory features. We began by developing a pipeline to extract and map components such as sound energy, spectral flux, fundamental frequency, and 13 MFCCs from the audio onto the accompanying brain data of participants. During the temporal alignment of the Tunnel audio, we noted a significant deviation in the HRF with a cross-correlation model that measured peak correlation to be a TR ahead of the dataset's alignment. After shifting to account for this lag, which may be an important characteristic of neuro-auditory response to consider in future studies, we measured how the brain encoded these features. Particularly, there was high correlation and feature weight from sonic energy, confirming previous studies, and also spectral flux, but negative weights for certain mid-level MFCCs.

Additionally, we constructed labels for the Tunnel audio that detailed whether at a given moment, if speech was heard or music. This enabled analysis of the brain's differential response to the two forms of auditory stimuli, and we found that with fair sampling procedures, there was a distinct ability to decode the brain's processing of either category. Though other studies go further into the finer details of how the brain encodes spectral

structure in more controlled settings through things like STRFs, this demonstrates how in naturalistic listening contexts when music and speech are placed and performed in conjunction with each other, the brain still distinguishes between the two. Future work may involve finding a stimulus with better balanced classes, as this study suffered from almost a 1:9 imbalance between the two classes, and that may improve findings. This might also make results such as measuring group synchronization during the two conditions more illustrative or significant.

We finished by measuring inter-subject correlation to capture the concept of synchronicity and heightened attention across all participants. Sliding window ISC revealed that spikes in global sound energy significantly boosted ISC as compared to baseline energy, demonstrating that participants may enter into shared neural states in these moments of drama and excitement. There still remains a lot of work to be done in measuring the effect of all the other features individually in encoding and decoding. The hemispheric asymmetry hypothesis is also one interesting way to continue this work, and see if speech and music are differentially encoded across right/left sides of the brain.

SCRIPT FOR SAVING FILES

So you don't have to wait 20 minutes to compute them again...

```
Saved final feature matrix.  
Saved raw inaSpeechSegmenter tags.  
Saved cleaned Gentle speech tags.
```

AI Appendix

A lot of the sections of my final project were things that were covered in previous labs, and therefore when it came to a lot of concepts and code blocks, I had no problem writing them on my own based off my work from the past. However, I did need to use AI support from Gemini Pro in the following areas:

- Broad support on structuring the notebook and the outline of analyses I wanted to cover
 - I uploaded my project proposal and a brainstorming document where I had thought of questions and specific analyses I wanted to conduct. I then asked it to help me structure a timeline for the project, the best layout of the topics, and any other ideas for things I could look at. This gave it context for the class and what I wanted to get done as a bare minimum.
- Preprocessing and audio feature extraction
 - I've never used Librosa or inaSpeechSegmenter before. It basically wrote all of the code for extracting individual features out of the audio file. I cross-referenced the

functions it called with the API documents from each library and made sure it was using them correctly, and crucially not manipulating any tuning parameters such as sampling rate or probabilistic Yin that I didn't understand or approve of.

- It also helped me create the cross-correlation plot in my early stages of the project, which was instrumental since I had accidentally over-corrected the plot by 2 TRs (not realizing the dataset already came aligned), but also helped uncover the HRF issue with the auditory cortex responding an entire TR faster than expected!
- The first Ridge Regression and GridSearch and ISC code block
 - I was running out of time and decided to have it write them, using some new functions from sklearn that I hadn't used before so I checked how those worked. I then copied them for the other analyses and replaced variable names as needed.
- Visualizations
 - I would put boilerplate code that I wrote for visualizations, or sometimes just the names of variables I had and what type of graph I wanted them on, and have the engine spit out the code for plots. This was mainly so the plots would look more aesthetic, after I had set up my basic x-y line graph plot.
 - This is how I got the confusion matrix and the idea to use the classification report.
- Any more custom analyses we haven't done in class
 - This includes either of the find peak experiments, where Gemini advised me on what parameters to choose and how to convolve them later with the sliding windows I had set up for ISC.
 - This also includes the t test for the peak experiments.
- I tried using it for debugging the permutation_isc experiment which had very irregular p-values that didn't match the graph. However all it came up with was that I was unpacking the function output wrong, which was untrue based off of brainiak's website documentation. So it was unhelpful there.

References

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